

Question number	Answer	Additional guidance	Mark
5(a)	An explanation that includes the following points: - because the mutated gene results in a faulty (CFTR) protein (1) - so chloride ions do not move out of the cells (1) - decreasing the water potential inside the cell (1) - therefore water {leaves the mucus / enters the cells} by <u>osmosis</u> (1)	ACCEPT less effective ACCEPT reduced movement of chloride ions / sodium ion channel not inhibited / sodium ions not prevented from entering cell ACCEPT decreasing {osmotic potential / solute potential} / increasing solute concentration IGNORE decreasing water concentration	(3)

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5(b)	An explanation that includes five of the following points: • because the mucus blocks the airways (1) • therefore {air flow to lungs / gas exchange} is reduced (1) • because mucus prevents <u>pancreatic</u> enzymes from entering the small intestine (1) • therefore large food molecules not broken down (and cannot be absorbed) (1) • because the mucus prevents sperm passing through the cervix (1) • therefore sperm cannot reach the egg cell (1)	ACCEPT duodenum / gut / digestive tract named enzyme secreted by pancreas ACCEPT named large food molecule ACCEPT reach the {oviducts / fallopian tubes}	(5)

Question number	Answer	Mark
5(c)(i)	C 40 The only correct answer is C. <i>A is incorrect because</i> <i>B is incorrect because</i> <i>D is incorrect because</i>	(1)

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5(c)(ii)	An explanation that includes the following points: - because (the graph shows that) the level of CFTR protein function varies between 3 and 18% (1) - because (the graph shows that) individuals diagnosed with cystic fibrosis have a range in concentration of chloride ions in sweat (1) - therefore the CFTR protein must be affected to different extents (by different mutations) (1)		(2)